Benchmarking Stochastic Optimization Algorithms for Non-Convex Objective Landscapes

Randomized Hill Climbing, RHCR5, Random Search, and Simulated Annealing

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Artificial Intelligence & Optimization
Python Implementation and Experimental Benchmarking

This project evaluates stochastic optimization algorithms on difficult non-convex benchmark functions. A custom five-stage randomized hill climbing method, RHCR5, is compared against Basic Randomized Hill Climbing, Random Search, and Simulated Annealing using repeated trials, convergence analysis, runtime comparison, success rates, and statistical significance testing.

Topics:	Stochastic Optimization, Local Search, AI Search Algorithms
Language:	Python
Algorithms:	Basic RHC, RHCR5, Random Search, Simulated Annealing
Benchmark Functions:	Frog, Rastrigin, Ackley
Trials:	50 random seeds per algorithm per function
Total Experiments:	600 benchmark trials
Date:	January 2026

Contents

1 Introduction

Optimization is the process of finding the best possible solution to a problem from a set of available choices. Many real-world tasks can be viewed as optimization problems. For example, a company may want to maximize profit, an engineer may want to minimize the weight of a structure while maintaining safety requirements, and a machine learning model may seek to minimize prediction error. In each case, the goal is to identify the input values that produce the most desirable outcome according to a mathematical objective function.

In simple optimization problems, it is often possible to calculate the best solution directly using algebra or calculus. However, many practical problems are much more complex. Their objective functions may contain thousands of possible solutions, irregular behavior, or many local optima. A local optimum is a solution that appears to be the best within a nearby region but is not necessarily the best solution overall. When an optimization landscape contains many local optima, traditional deterministic methods can become trapped and fail to locate higher-quality solutions elsewhere in the search space.

To address these challenges, researchers often use stochastic optimization algorithms. The term *stochastic* refers to the use of randomness during the search process. Rather than following a fixed path, stochastic methods explore the search space by generating and evaluating candidate solutions in a probabilistic manner. This randomness can help the algorithm avoid poor local optima and discover regions that might otherwise be overlooked. As a result, stochastic optimization techniques have become widely used in artificial intelligence, machine learning, operations research, engineering design, and scientific computing.

This study investigates the performance of four stochastic optimization methods: Basic Randomized Hill Climbing (RHC), Random Search, Simulated Annealing (SA), and a custom algorithm called Randomized Hill Climbing with Five-Stage Refinement (RHCR5). Each method approaches the optimization process differently. Random Search samples solutions from the search space without considering previous results, Randomized Hill Climbing repeatedly moves toward better nearby solutions, and Simulated Annealing occasionally accepts worse solutions in order to escape local optima. RHCR5 extends hill climbing by combining multiple stages of progressively smaller search neighborhoods, allowing it to transition from broad exploration to precise local refinement.

To evaluate these algorithms, three benchmark functions were selected: the Frog function, the Rastrigin function, and the Ackley function. These functions are commonly used in optimization research because they contain complex landscapes with many local optima and varying levels of difficulty. By testing algorithms on benchmark functions with different characteristics, it becomes possible to compare how effectively each method balances exploration of the search space with exploitation of promising solutions.

The primary objective of this study is to compare the effectiveness, efficiency, and reliability of the four optimization methods. Performance is evaluated through repeated randomized trials and measured using solution quality, convergence behavior, computational cost, runtime, success rate, and statistical significance testing. By examining both the best solutions found and the average performance across many independent runs, the study aims to provide a comprehensive assessment of each algorithm's strengths and weaknesses.

Through systematic experimentation and analysis, this project demonstrates how stochastic optimization algorithms behave on difficult non-convex problems and explores whether a multi-stage refinement strategy such as RHCR5 can improve optimization performance across a variety of objective landscapes.

2 Benchmark Functions

2.1 Frog Function

The Frog function is defined as:

$$f(x, y) = x\cos(\sqrt{|x + y + 1|})\sin(\sqrt{|y - x + 1|}) + (1 + y)\sin(\sqrt{|x + y + 1|})\cos(\sqrt{|y - x + 1|})$$

with domain:

$$(x, y) \in [-512, 512] \times [-512, 512].$$

This function is difficult because it contains many local minima and highly irregular behavior. The best known target value used in this project is approximately:

$$f(x, y) \approx -512.$$

2.2 Rastrigin Function

The Rastrigin function is defined as:

$$f(x, y) = 20 + x^2 + y^2 - 10\cos(2\pi x) - 10\cos(2\pi y)$$

with domain:

$$(x, y) \in [-5.12, 5.12] \times [-5.12, 5.12].$$

The global minimum is:

$$f(0, 0) = 0.$$

The Rastrigin function is commonly used in optimization because it has many local minima surrounding the global minimum.

2.3 Ackley Function

The Ackley function is defined as:

$$f(x, y) = -20e^{-0.2\sqrt{0.5(x^2+y^2)}} - e^{0.5(\cos(2\pi x)+\cos(2\pi y))} + e + 20.$$

with domain:

$$(x, y) \in [-32.768, 32.768] \times [-32.768, 32.768].$$

The global minimum is:

$$f(0, 0) = 0.$$

Ackley is another standard non-convex optimization benchmark. It has a broad outer region and a narrow basin near the global minimum, making it useful for testing whether an algorithm can balance exploration and refinement.

3 Algorithms

3.1 Basic Randomized Hill Climbing

Basic Randomized Hill Climbing begins from a starting point and samples random neighboring points within a square neighborhood of size z. At each iteration, the algorithm evaluates p neighbors. If any neighbor improves the objective value, the algorithm moves to the best sampled neighbor. If no improvement is found, the algorithm stops.

This method is simple and efficient, but it can easily become trapped in local minima.

3.2 RHCR5

RHCR5 is the custom algorithm developed for this project. It extends randomized hill climbing by applying five stages of progressively smaller neighborhoods. The stages use neighborhood sizes:

$$z, \frac{z}{10}, \frac{z}{50}, \frac{z}{250}, \frac{z}{1000}.$$

The solution from one stage becomes the starting point for the next stage. The purpose of this design is to first explore broadly and then refine the best region more precisely.

This structure allows RHCR5 to behave like a coarse-to-fine optimization method. The early stages help escape weak regions, while the later stages improve precision around promising minima.

3.3 Random Search

Random Search samples points uniformly from the full search domain and keeps the best solution found. It is a useful baseline because it has no local search behavior and does not use previous points to guide future search.

Although simple, Random Search can sometimes perform surprisingly well in low-dimensional spaces.

3.4 Simulated Annealing

Simulated Annealing is a probabilistic search algorithm that sometimes accepts worse solutions to escape local minima. The probability of accepting a worse move decreases as the temperature cools.

This makes Simulated Annealing a strong comparison method because it balances exploration and exploitation.

4 Experimental Setup

Each algorithm was tested on each benchmark function using 50 random seeds. This produced:

$$4 \text{ algorithms} \times 3 \text{ functions} \times 50 \text{ seeds} = 600 \text{ total trials.}$$

Each run was evaluated using the following metrics:

- final objective value
- best observed value
- mean objective value
- standard deviation
- median objective value
- function calls
- runtime in seconds
- success rate
- error from known target value
- statistical significance

For the Frog function, success was defined as achieving:

$$f(x, y) \leq -510.$$

For Rastrigin and Ackley, success was defined as achieving:

$$f(x, y) \leq 1.$$

5 Best Results

Table 1: Best observed result for each benchmark function

Function	Best Algorithm	Best Value	x	y	Calls
Frog	RHCR5	-511.732881886610	-488.632579	512.000000	6005
Rastrigin	RHCR5	0.000000030692	0.000011	-0.000005	5105
Ackley	RHCR5	0.000029645737	-0.000003	0.000010	4205

RHCR5 produced the best observed solution on all three benchmark functions. On the Frog function, the best value was:

$$-511.732881886610.$$

Using the approximate target of -512, the error was:

$$0.267118.$$

The percent error was:

$$0.052172\%.$$

This means RHCR5 reached a solution extremely close to the best-known target value for the Frog function.

6 Full Benchmark Summary

Table 2: Benchmark summary for the Ackley function

Algorithm	Best	Mean	Std. Dev.	Mean Calls	Success Rate
Basic RHC	0.172287	2.104657	0.905011	1465	14%
RHCR5	0.000030	0.000790	0.000460	4571	100%
Random Search	0.131891	2.458788	0.929002	7203	12%
Simulated Annealing	0.038387	0.561362	0.415072	7203	84%

For Ackley, RHCR5 was clearly the strongest method. It achieved the best value, the lowest mean value, the lowest standard deviation, and a 100% success rate.

Table 3: Benchmark summary for the Frog function

Algorithm	Best	Mean	Std. Dev.	Mean Calls	Success Rate
Basic RHC	-511.728551	-499.671004	21.782114	3325	50%
RHCR5	-511.732882	-500.555626	21.952347	7107	56%
Random Search	-510.592918	-502.145020	5.874606	7203	6%
Simulated Annealing	-511.732876	-502.564738	12.062102	7203	48%

For the Frog function, RHCR5 found the best observed solution and had the highest success rate. However, the differences between RHCR5, Basic RHC, Random Search, and Simulated Annealing were not statistically significant. This shows that the Frog function is highly sensitive to initialization and that several stochastic methods can reach strong local minima.

Table 4: Benchmark summary for the Rastrigin function

Algorithm	Best	Mean	Std. Dev.	Mean Calls	Success Rate
Basic RHC	0.000704	0.628789	0.543457	1771	78%
RHCR5	0.000000031	0.198993	0.449477	4787	98%
Random Search	0.005976	0.692860	0.390136	7203	72%
Simulated Annealing	0.004118	0.082005	0.079020	7203	100%

For Rastrigin, RHCR5 found the best individual solution and had a strong 98% success rate. However, Simulated Annealing had the best average performance and a 100% success rate. This suggests that Simulated Annealing was more consistent on Rastrigin, while RHCR5 was capable of finding the closest individual solution to the global minimum.

7 Statistical Significance

Welch's two-sample t-tests were used to compare RHCR5 against the other algorithms. The null hypothesis was that RHCR5 and the comparison algorithm had equal mean final objective values.

Table 5: Statistical significance tests comparing RHCR5 against other algorithms

Function	Comparison	t-statistic	p-value	Significant?
Frog	RHCR5 vs Basic RHC	-0.202270	0.840125	No
Frog	RHCR5 vs Random Search	0.494557	0.622848	No
Frog	RHCR5 vs Simulated Annealing	0.567175	0.572264	No
Rastrigin	RHCR5 vs Basic RHC	-4.309287	0.000040	Yes
Rastrigin	RHCR5 vs Random Search	-5.867446	0.000000063	Yes
Rastrigin	RHCR5 vs Simulated Annealing	1.812625	0.075661	No
Ackley	RHCR5 vs Basic RHC	-16.438014	1.41×10^{-21}	Yes
Ackley	RHCR5 vs Random Search	-18.708975	5.85×10^{-24}	Yes
Ackley	RHCR5 vs Simulated Annealing	-9.549761	9.08×10^{-13}	Yes

The statistical results show that RHCR5 was significantly better than all other methods on Ackley. On Rastrigin, RHCR5 significantly outperformed Basic RHC and Random Search, but not Simulated Annealing. On the Frog function, the differences were not statistically significant.

8 Visual Results

8.1 Boxplots

The boxplots compare the distribution of final objective values across 50 random seeds.

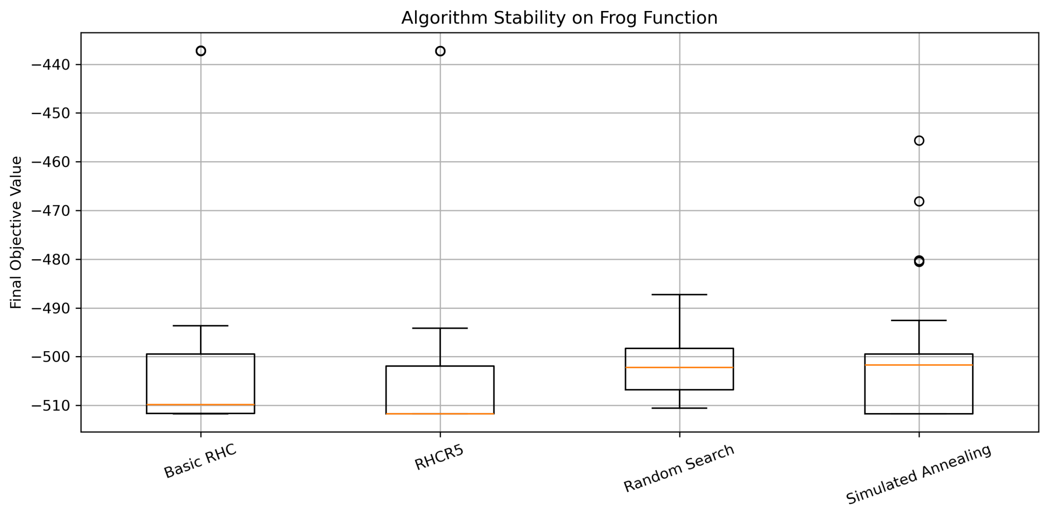

Figure 1: Algorithm stability on the Frog function across 50 random seeds.

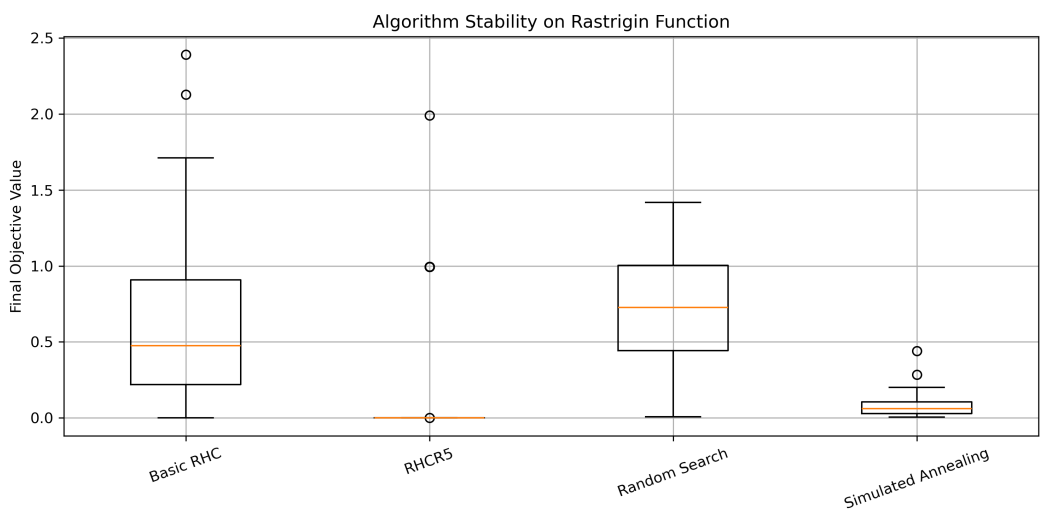

Figure 2: Algorithm stability on the Rastrigin function across 50 random seeds.

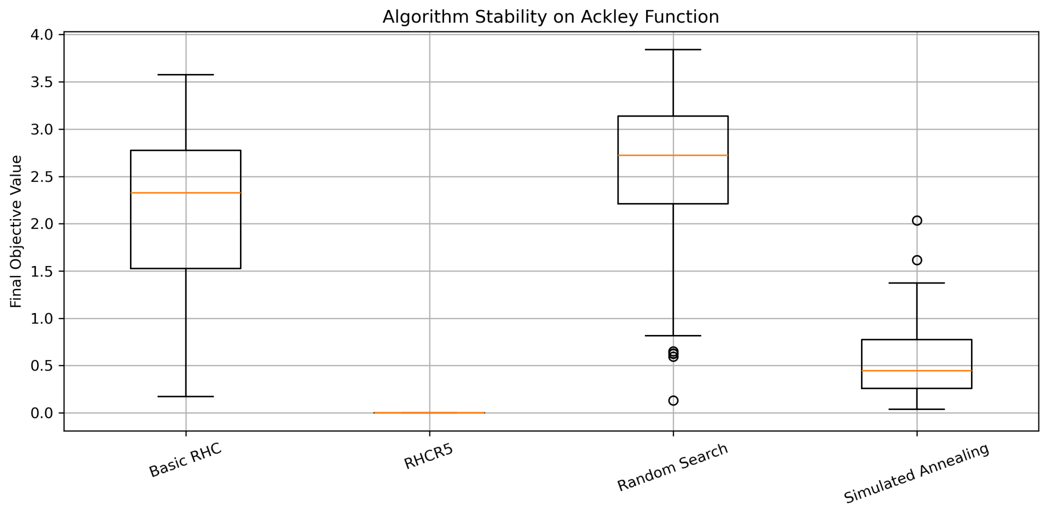

Figure 3: Algorithm stability on the Ackley function across 50 random seeds.

8.2 Runtime Charts

Runtime charts show the average execution time of each algorithm.

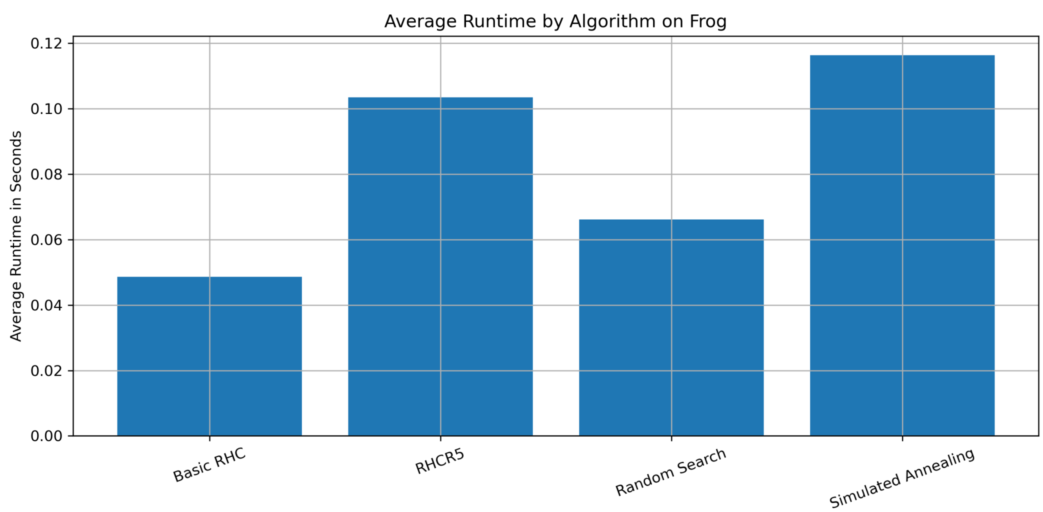

Figure 4: Average runtime by algorithm on the Frog function.

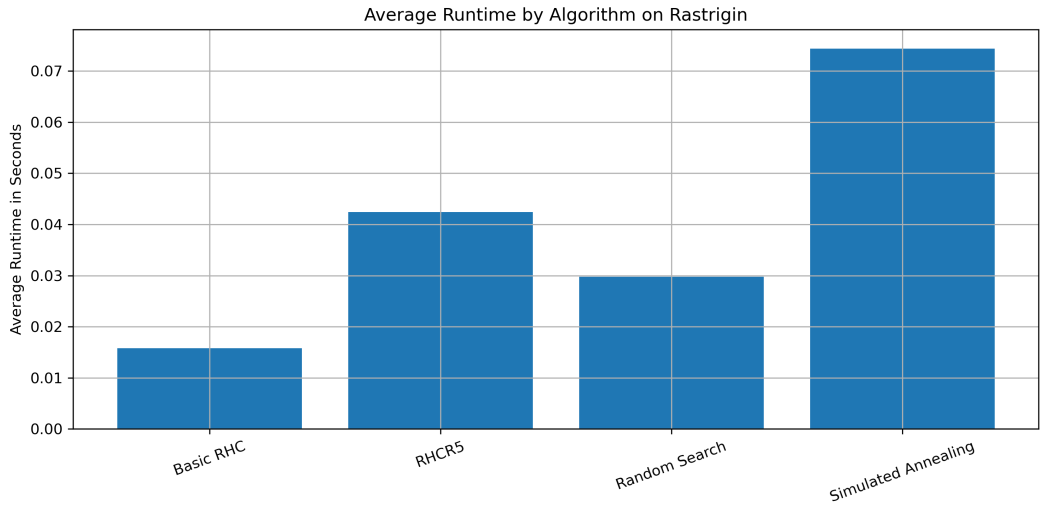

Figure 5: Average runtime by algorithm on the Rastrigin function.

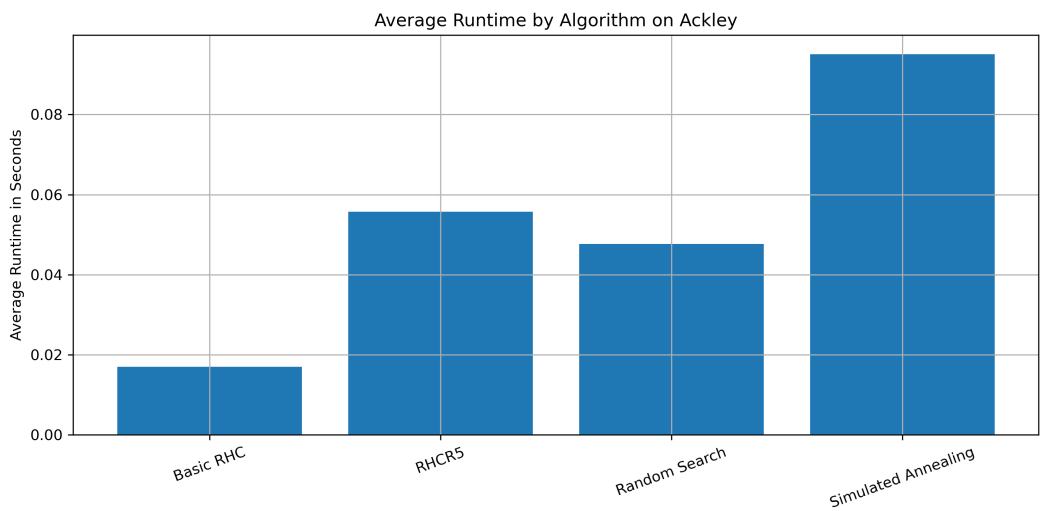

Figure 6: Average runtime by algorithm on the Ackley function.

8.3 Success Rate Charts

Success rate measures how often each algorithm reached the chosen success threshold.

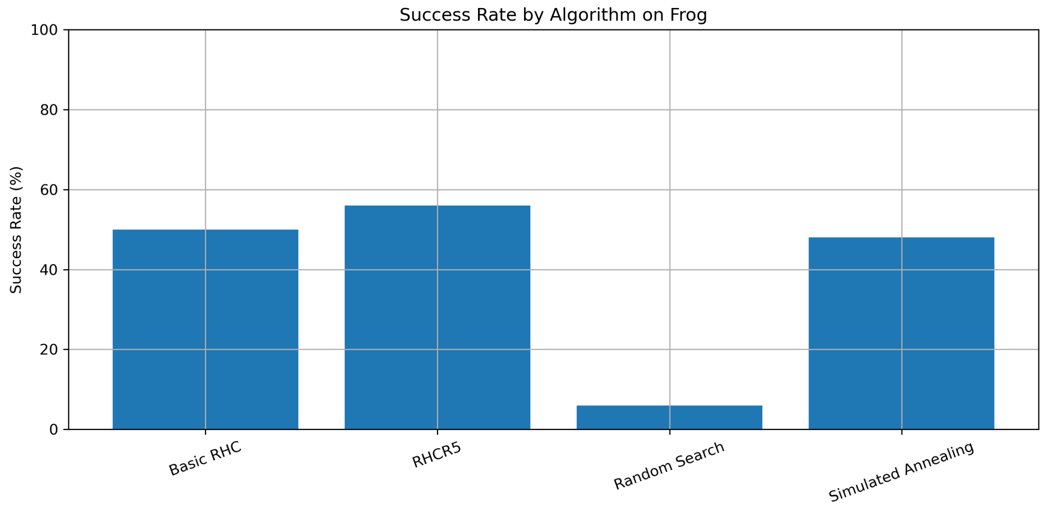

Figure 7: Success rate by algorithm on the Frog function. Success was defined as $f(x, y) \leq -510$.

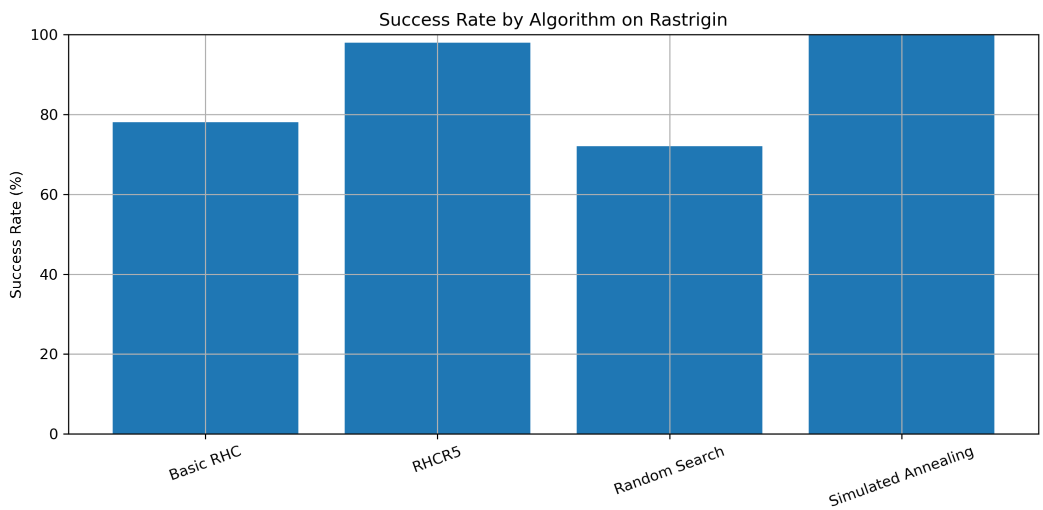

Figure 8: Success rate by algorithm on the Rastrigin function. Success was defined as $f(x, y) \leq 1$.

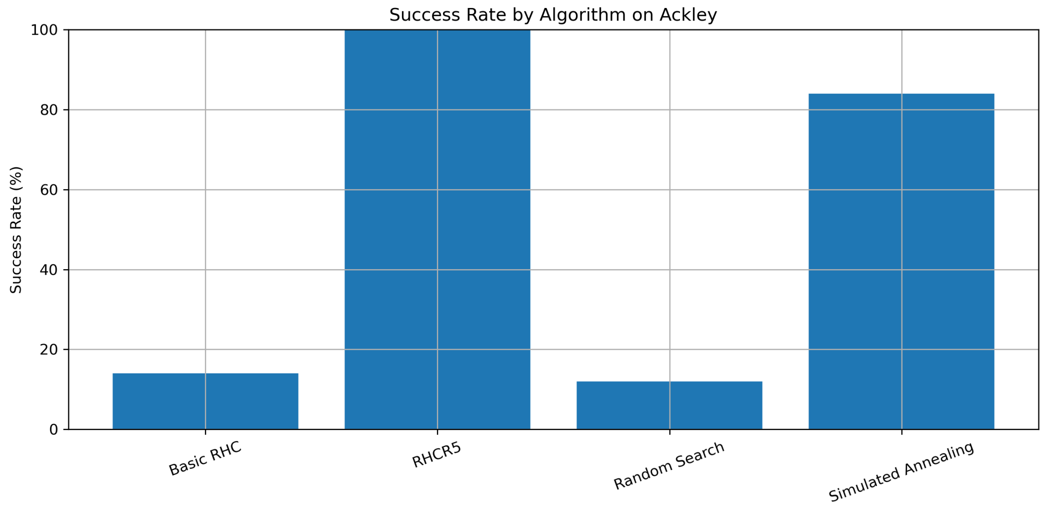

Figure 9: Success rate by algorithm on the Ackley function. Success was defined as $f(x, y) \leq 1$.

8.4 Mean Convergence Curves

The convergence plots show the mean best objective value over time across 50 random seeds.

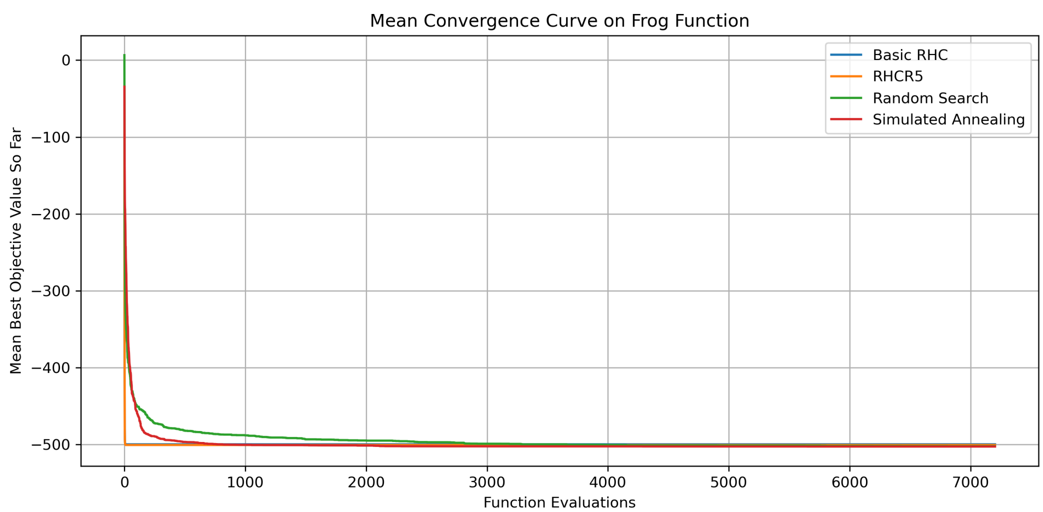

Figure 10: Mean convergence curve on the Frog function.

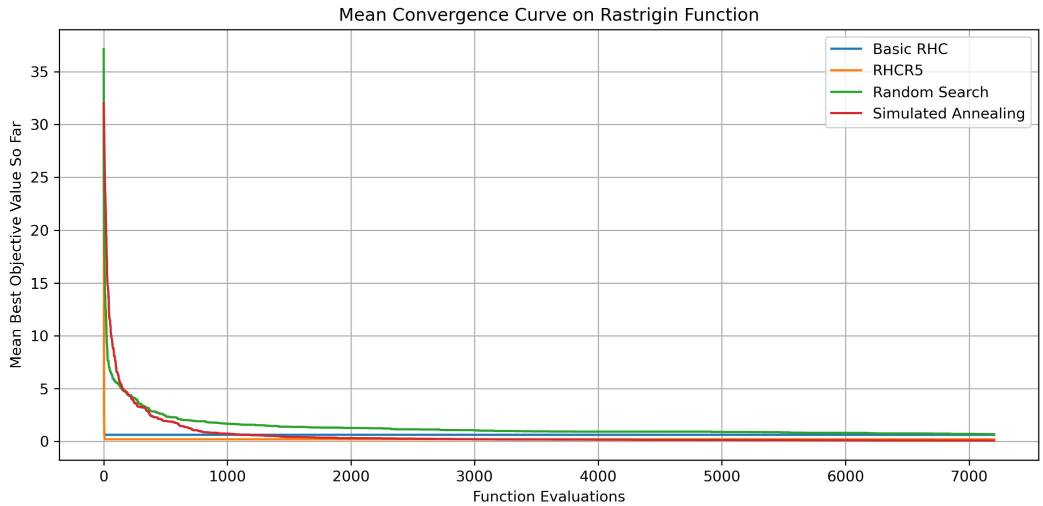

Figure 11: Mean convergence curve on the Rastrigin function.

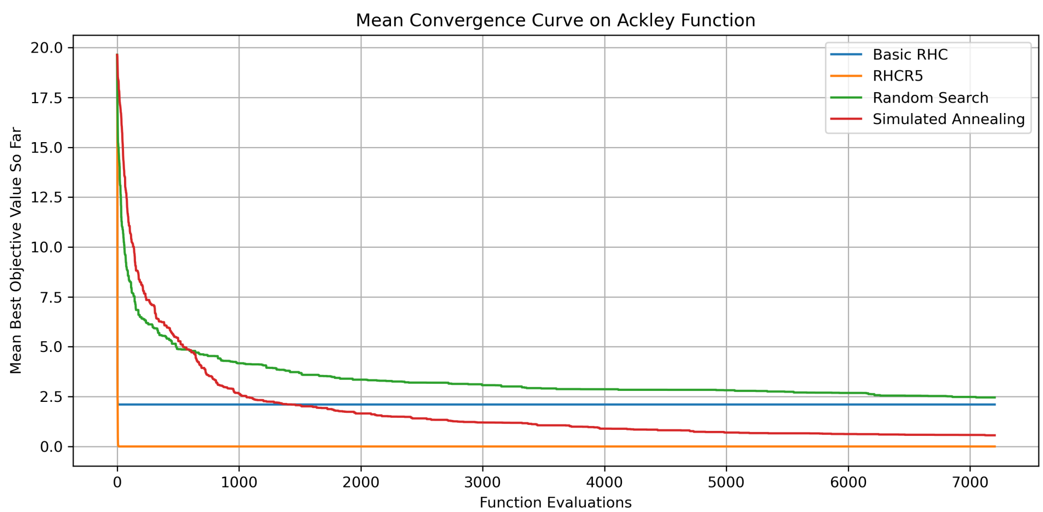

Figure 12: Mean convergence curve on the Ackley function.

8.5 Function Landscapes and RHCR5 Search Paths

The following heatmaps show the objective-function landscape and the RHCR5 trajectory.

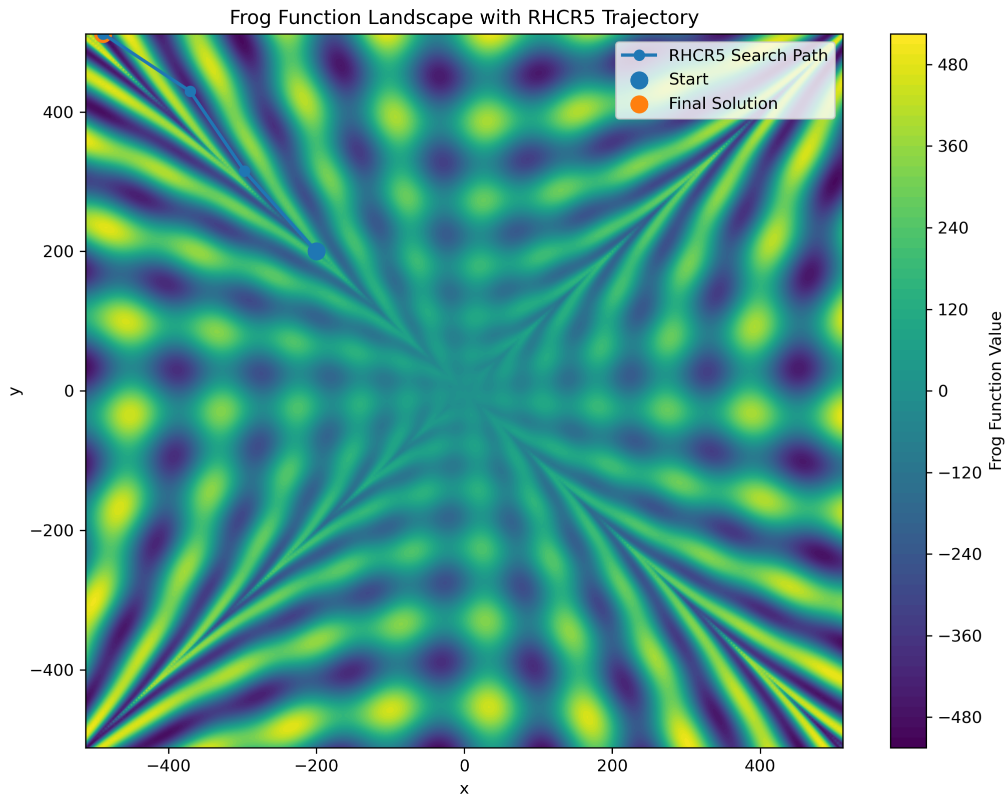

Figure 13: Frog function landscape with RHCR5 search trajectory.

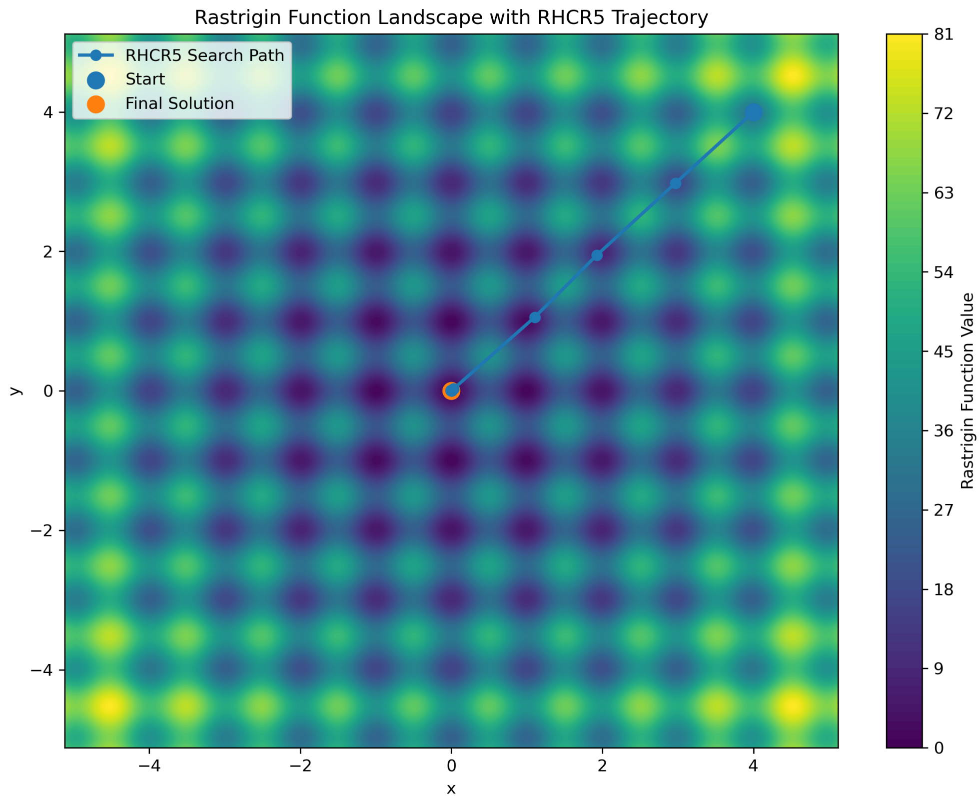

Figure 14: Rastrigin function landscape with RHCR5 search trajectory.

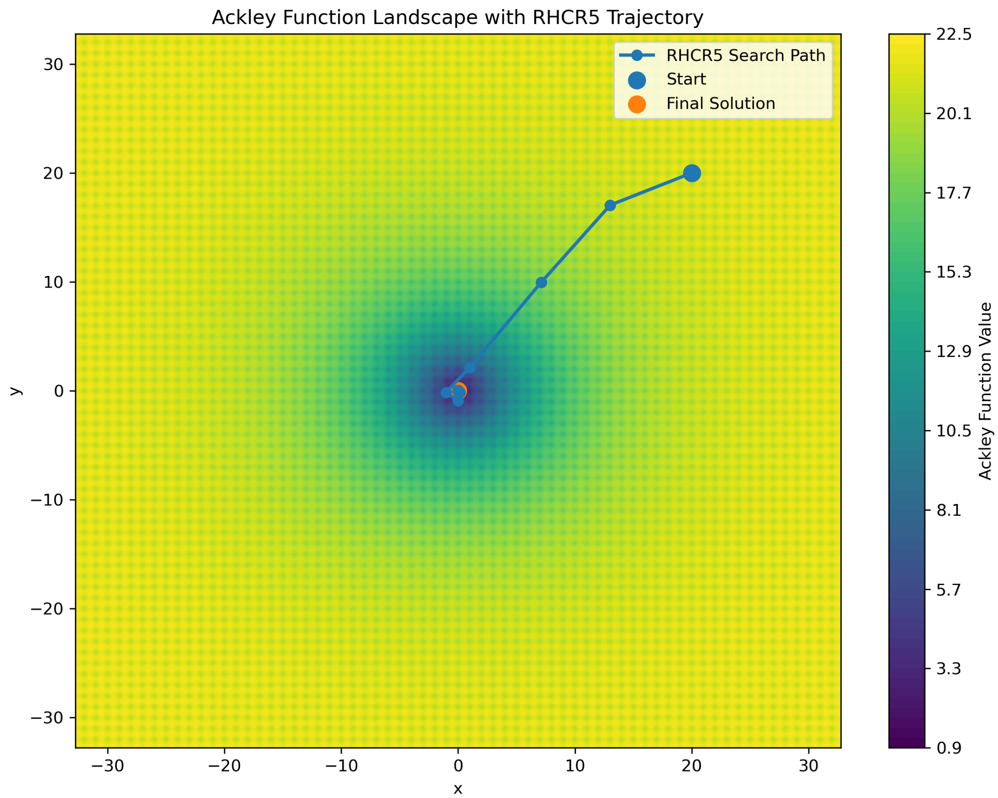

Figure 15: Ackley function landscape with RHCR5 search trajectory.

9 Discussion

Before interpreting the results, it is important to understand what a successful optimization algorithm is trying to accomplish. The goal of an optimization algorithm is to search through a large number of possible solutions and identify one that produces the best objective value. For difficult problems, there is no guarantee that an algorithm will find the global optimum, so performance is often evaluated based on how close the algorithm gets to the best-known solution and how consistently it can achieve that result across repeated trials.

The results of this study show that RHCR5 performed strongly across all three benchmark functions. In every case, it produced the best solution observed during the experiments. This is particularly noteworthy because RHCR5 does not use gradients, derivatives, or any detailed information about the shape of the objective function. Instead, it relies entirely on evaluating candidate solutions and gradually improving them through a sequence of increasingly refined searches.

The strongest performance was observed on the Ackley function. RHCR5 achieved a best objective value of approximately

[0.000029645737,]

which is extremely close to the true optimum value of 0. In addition, the algorithm achieved a 100

RHCR5 also performed very well on the Rastrigin function. The best solution found was

[0.000000030692,]

which is essentially indistinguishable from the true global optimum of 0. The algorithm achieved a 98

The most difficult benchmark in this study was the Frog function. Unlike Ackley and Rastrigin, the Frog function contains a highly irregular landscape with many competing regions that produce similar objective values. RHCR5 found the best solution observed during the experiments,

[-511.732881886610,]

which is approximately 0.052

One of the most important observations from this study is the role of the multi-stage refinement strategy. Traditional hill climbing typically uses a single neighborhood size throughout the search process. In contrast, RHCR5 begins with large neighborhoods that encourage exploration of the search space and then progressively reduces the neighborhood size to improve precision. This approach mirrors how a person might search for a lost object: first scanning a large area and then focusing more carefully on the most promising locations. The experimental results suggest that this combination of exploration and refinement contributes significantly to the algorithm's effectiveness.

Overall, RHCR5 demonstrated strong performance across a variety of optimization landscapes. It consistently produced high-quality solutions, achieved the best observed result on every benchmark function, and showed particular strength on Ackley and Rastrigin. These findings indicate that multi-stage local search can be an effective strategy for solving difficult optimization problems.

10 Limitations

Although the results are encouraging, several limitations should be considered when interpreting the findings.

First, all experiments were performed using two-dimensional benchmark functions. These functions are useful because they allow algorithms to be compared in a controlled environment, but many real-world optimization problems involve dozens, hundreds, or even thousands of variables. An algorithm that performs well in two dimensions may behave differently as the size and complexity of the problem increase. Additional experiments would therefore be needed to evaluate how RHCR5 scales to higher-dimensional optimization tasks.

Second, all algorithms were given the same budget of function evaluations. This creates a fair comparison because each method receives the same amount of computational effort. However, different algorithms may benefit differently from larger or smaller budgets. Future work could investigate how performance changes as the number of allowed evaluations increases.

Third, the parameter values used in this study were selected through experimentation but were not exhaustively optimized. Parameters such as neighborhood size, cooling schedules, stopping criteria, and refinement rates can have a substantial impact on performance. It is possible that additional tuning could improve the results for one or more algorithms.

Finally, the Frog function results illustrate an important limitation of experimental benchmarking. Although RHCR5 found the best observed solution, the statistical tests did not show a significant difference between RHCR5 and the competing methods. This indicates that some optimization landscapes may be difficult to distinguish using a finite number of trials. Conducting additional experiments with more random seeds or alternative parameter settings could provide a clearer picture of the relative strengths of the algorithms on particularly challenging problems.

Despite these limitations, the benchmark provides a useful comparison of several stochastic optimization methods and demonstrates the potential benefits of the RHCR5 approach.

11 Conclusion

Optimization algorithms play an important role in many areas of science, engineering, and artificial intelligence because they provide a systematic way to search for high-quality solutions to difficult problems. This project investigated four stochastic optimization methods and introduced RHCR5, a custom algorithm that combines randomized hill climbing with a five-stage refinement process.

The algorithms were evaluated on three benchmark functions—Frog, Rastrigin, and Ackley—using a total of 600 experimental trials. Across all benchmark functions, RHCR5 produced the best observed solution. The algorithm achieved a 100

Perhaps the most important finding is that combining broad exploration with increasingly precise local search can be highly effective. By beginning with larger search neighborhoods and gradually reducing their size, RHCR5 is able to first identify promising regions of the search space and then refine those solutions with greater accuracy. The results suggest that this strategy can improve optimization performance across a variety of objective landscapes.

Beyond the specific algorithm developed in this study, the project demonstrates the broader process of algorithm benchmarking. Through repeated trials, statistical analysis, visualization, and comparative evaluation, it is possible to gain meaningful insight into the strengths and weaknesses of different optimization methods. The findings provide evidence that RHCR5 is a robust and competitive optimization algorithm and establish a foundation for future research involving higher-dimensional problems, additional benchmark functions, and more advanced optimization techniques.